A Bin-Packing Mean-Field Analysis

This appendix derives two expected-value results for the thresholded reinsertion policy of §4.2.1 under mean-field assumptions: the characteristic time between local repairs, and the steady-state active bin count.

A.1 Model and Assumptions

Let each model demand vector be $v_i \in [0,1]^3$, and let each bin have normalized capacity $(1,1,1)$. For model i, define its scalar size $s_i \triangleq \|v_i\|_1/3 \in [0,1]$ and write $v_i = s_i\mathbf{1} + \xi_i$ with $\langle \xi_i, \mathbf{1}\rangle = 0$. Then $\Phi(v_i) = \|\xi_i\|_2^2$. Let $\bar{s} \triangleq \mathbb{E}[s_i]$ and $\bar{\phi} \triangleq \mathbb{E}[\Phi(v_i)]$. For a bin B, let $r_B = \mathbf{1} - \sum_{i\in B} v_i$ be the residual vector, and define the damage ratio $\rho(B) \triangleq \Phi(r_B)/\sum_{i\in B}\Phi(v_i)$.

We use the following assumptions:

(A1) Model arrivals form a Poisson process with rate λ; each active model departs independently after an exponential lifetime with rate μ, giving $\mathbb{E}[N_{\text{act}}] = \lambda/\mu$.

(A2) The workload distribution is non-adversarial: $s_i \in [0,1]$, $\bar{s} \in (0,1)$, $\bar{\phi} \in (0,\infty)$.

(A3) Immediately after a local repack or global consolidation, a touched bin has expected damage ratio $\rho_{\text{reset}} \in (0,1)$.

(A4) The characteristic local threshold-crossing time is approximated by the first time at which the departure-only proxy trajectory reaches α.

(A5) At stationarity, if a random active bin B has residual r_B, then $\delta_B \triangleq \min_d r_{B,d}$ has approximately constant mean $\eta \triangleq \mathbb{E}[\delta_B] \in [0,1)$.

A.2 Repacking Frequency

Theorem A.1 (Characteristic repair time)**.** *Under Assumptions A1, A3, and A4, the characteristic local threshold-crossing time satisfies*

$$\tau_\alpha \approx \frac{1}{\mu}\log\!\left(\frac{1-\rho_{\text{reset}}}{1-\alpha}\right).$$

Proof. Fix a tagged bin immediately after a local repair, and let B_0 denote its resident set. Define $A_0 \triangleq \sum_{i\in B_0}\Phi(v_i)$ and $C_0 \triangleq \sum_{i\neq j\in B_0}\langle \xi_i, \xi_j\rangle$. Since $\Phi\big(\sum_{i\in B_0}\xi_i\big) = A_0 + C_0$, the initial damage ratio is $\rho_{\text{reset}} = 1 + C_0/A_0$. Because $\rho_{\text{reset}} < 1$, we have $C_0 < 0$, reflecting the complementarity created by repair.

Ignoring future arrivals and tracking only the surviving subset under Assumption A1, each model survives to time t with probability $q(t) = e^{-\mu t}$. The intrinsic and cross terms thin at different rates: $\mathbb{E}[A_t \mid B_0] = q(t)A_0$ and $\mathbb{E}[C_t \mid B_0] = q(t)^2 C_0$. The departure-only proxy damage ratio is therefore

$$\tilde{\rho}(t) = 1 - e^{-\mu t}(1-\rho_{\text{reset}}).$$

Setting $\tilde{\rho}(t) = \alpha$ and solving gives the claimed expression. □

Corollary A.2 (Effective reset interval)**.** *If the background global consolidation runs every $T_g = 30$ s, the effective reset interval is $\tau_\alpha^{\text{eff}} \approx \min\{\tau_\alpha, T_g\}$.*

Since departures arrive at rate μ, the amortized repacking cost per model is $1/(\mu\tau_\alpha) \approx 1/\log\big((1-\rho_{\text{reset}})/(1-\alpha)\big)$, an $O(1)$ constant confirming bounded migration churn.

A.3 Steady-State Bin Count

Theorem A.3 (Bin-count ratio)**.** *Under Assumptions A1, A2, and A5, together with the mean-field closure induced by the local threshold rule, the steady-state overhead ratio $\mathcal{R}_{\text{mf}} \triangleq \mathbb{E}[N]/U_0$ where $U_0 = \lambda\bar{s}/\mu$ is approximated by*

$$\mathcal{R}_{\text{mf}} \approx \left(\frac{\beta\sqrt{\zeta} + \sqrt{\beta^2\zeta + 4(1-\eta)}}{2(1-\eta)}\right)^2,$$

where $\beta \triangleq \sqrt{2\alpha/3}$ and $\zeta \triangleq \bar{\phi}/\bar{s}$.

Proof. Let $U = \sum_i s_i$ be the total normalized scalar volume and N the number of active bins. For each active bin B, define $\delta_B = \min_d r_{B,d}$ and $F_B = \|r_B\|_1 - 3\delta_B$. The scalar-capacity accounting identity is $N = U + \sum_B \delta_B + \frac{1}{3}\sum_B F_B$. Let $y_B = r_B - \delta_B\mathbf{1}$; one coordinate is zero and the other two are nonneg., giving $F_B = u_B + v_B$. Since Φ is shift-invariant along $\mathbf{1}$, $\Phi(r_B) = \Phi(y_B) \geq \frac{1}{6}(u_B+v_B)^2 = \frac{1}{6}F_B^2$, so $F_B \leq \sqrt{6\,\Phi(r_B)}$.

Define $A_B = \sum_{i\in B}\Phi(v_i)$ and $A = \sum_B A_B$. Under the local threshold rule, $\rho(B) \lesssim \alpha$, so $F_B \lesssim \sqrt{6\alpha\, A_B}$. Summing over bins and applying Cauchy–Schwarz: $\sum_B F_B \lesssim \sqrt{6\alpha\, N\, A}$.

Taking expectations in the accounting identity above with $\mathbb{E}[U] = U_0$ and $\mathbb{E}[A] = \lambda\bar{\phi}/\mu$, letting $x = \sqrt{\mathbb{E}[N]}$, and solving the resulting quadratic $(1-\eta)x^2 - \beta\sqrt{A_0}\,x - U_0 \approx 0$ gives the claimed formula after dividing by U_0 and substituting $\zeta = \bar{\phi}/\bar{s}$. □

In the light-fragmentation regime (small η and $\beta\sqrt{\zeta}$), this simplifies to $\mathcal{R}_{\text{mf}} \approx (1-\eta)^{-1} + \beta\sqrt{\zeta}\,(1-\eta)^{-3/2}$, confirming near-unit overhead when bins are well packed.

B LST-IMH Algorithm

B.1 Pseudocode

Algorithm 1 LST-IMH: Latest-Start-Time Iterative Moore–Hodgson

Require: Jobs J with (p_j, d_j); machines with availability times $T_1 \geq \cdots \geq T_m$.
Ensure: Feasible schedule and on-time job set S.

$R \leftarrow J$; $S \leftarrow \emptyset$.
for $i = 1, 2, \ldots, m$ **do**
  $R_i \leftarrow \{j \in R : d_j - T_i \geq 0\}$; $d_j^{(i)} \leftarrow d_j - T_i$.
  $S_i \leftarrow$ Moore-Hodgson$(R_i, \{p_j\}, \{d_j^{(i)}\})$.
  Schedule S_i on M_i in EDD order from T_i.
  $S \leftarrow S \cup S_i$; $R \leftarrow R \setminus S_i$.
end for
return S.

Algorithm 2 Moore–Hodgson (single machine)

Require: Jobs R with processing times $\{p_j\}$ and deadlines $\{d_j\}$.
Ensure: Maximum-cardinality on-time job set.

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1:  Sort R by non-decreasing d_j (EDD order).
2:  S ← ∅;  t ← 0;  H ← empty max-heap keyed on p_j.
3:  for each j in EDD order do
4:     S ← S ∪ {j};  H.push(j);  t ← t + p_j.
5:     if t > d_j then
6:        j* ← H.pop() {largest p_j*}
7:        S ← S \ {j*};  t ← t − p_j*.
8:     end if
9:  end for
10: return S.
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B.2 Approximation Guarantee

Theorem B.1 (2-Approximation). *Let OPT be the maximum number of on-time jobs in an optimal schedule for $P_m \mid T_i \mid \sum U_j$, and let $ALG = |S|$ be the count returned by LST-IMH. Then $ALG \geq \frac{1}{2}$ OPT.*

Proof. Fix an optimal schedule and let O_i denote the set of jobs completed on time on machine M_i in this optimal solution. The sets $O_1, \ldots, O_m$ are pairwise disjoint and satisfy $\bigcup_{i=1}^{m} O_i = OPT$.

Consider iteration i of LST-IMH. The current residual job set is $R = J \setminus (S_1 \cup \cdots \cup S_{i-1})$. Among the jobs that the optimal solution assigns to M_i, those not yet claimed by earlier iterations form the subset

$$O_i \cap R \;=\; O_i \setminus (S_1 \cup \cdots \cup S_{i-1}).$$

Because the full set O_i is feasible on M_i starting at T_i (by optimality), removing some of its members can only relax the schedule: $O_i \cap R$ remains a feasible candidate for the single-machine sub-problem on M_i.

The algorithm invokes Moore–Hodgson on M_i, which returns a maximum-cardinality on-time subset for a single machine. Since $O_i \cap R$ is one feasible solution, we have

$$|S_i| \;\geq\; |O_i \cap R| \;=\; |O_i \setminus (S_1 \cup \cdots \cup S_{i-1})| \;\geq\; |O_i \setminus S|, \quad (1)$$

where the last inequality holds because $S \supseteq S_1 \cup \cdots \cup S_{i-1}$, so subtracting a larger set can only shrink the remainder.

Summing (1) over $i = 1, \ldots, m$:

$$|S| = \sum_{i=1}^{m} |S_i| \;\geq\; \sum_{i=1}^{m} |O_i \setminus S|.$$

Because the O_i are pairwise disjoint, so are the $O_i \setminus S$, giving

$$\sum_{i=1}^{m} |O_i \setminus S| \;=\; \left|\bigcup_{i=1}^{m} (O_i \setminus S)\right| \;=\; \left|\left(\bigcup_{i=1}^{m} O_i\right) \setminus S\right| \;=\; |OPT \setminus S| \;=\; |OPT| - |OPT \cap S|.$$

Combining with the trivial bound $|OPT \cap S| \leq |S|$ yields

$$|S| \;\geq\; |OPT| - |S| \qquad \Longrightarrow \qquad 2|S| \;\geq\; |OPT|. \qquad \square$$

B.3 Remarks

Remark 1 (Order independence). The set-theoretic relaxation $|O_i \setminus (S_1 \cup \cdots \cup S_{i-1})| \geq |O_i \setminus S|$ holds for any processing order of the machines. Consequently, the $\frac{1}{2}$-approximation guarantee is valid regardless of how machines are sorted.

Remark 2 (Why LST ordering). Although the bound is order-independent, the LST (Latest Start Time) ordering is crucial in practice. A machine with a large availability time T_i has a restricted feasible domain: only jobs satisfying $d_j - T_i \geq p_j$ can be completed on time. If lightly loaded machines (small T_i) are processed first, they greedily consume "easy" jobs—those with generous slack that *any* machine could handle—while leaving tight-deadline jobs for the constrained machines. This can exhaust the constrained machines' feasible sets entirely, pushing performance to the worst-case $\frac{1}{2}$ OPT boundary. Processing the most constrained machine first reverses this dynamic: it reserves easy jobs for flexible machines that can absorb them later, consistently yielding near-optimal solutions in practice.

C P/D Disaggregation Adaptation

Prefill–decode (P/D) disaggregation requires the decode node to pre-allocate KV cache blocks and register their addresses with the communication backend so that the prefill node can write KV data directly to the target location. Under virtual HBM management, however, a freshly allocated KV block has only a virtual address with no physical page bound, so the standard memory-registration call fails.

Janus solves this by having the decode node register the global xTensor with the communication backend at instance startup and expose its base address to the prefill node. When the decode node allocates a KV block, it looks up the block's physical page IDs and computes their offsets within the global xTensor. The prefill node then writes KV data to base address plus offsets, landing the data at the correct physical location without any per-block registration. The P/D framework logic is unchanged; only the address translation layer is adapted.

D Feature Selection for Capacity GP

Table 2 reports the normalized mutual information (NMI) between each input feature and the three resource dimensions on production traces. Across all three dimensions, token rate λ_t exhibits the highest NMI (0.63, 0.47, 0.48), confirming that instantaneous load is the dominant factor. Input length μ_{in} ranks second for every dimension (0.45, 0.42, 0.43): longer inputs inflate both KV cache footprint and prefill compute. Request rate λ_r contributes moderately (0.35, 0.36, 0.34), while output length and input variance have smaller but non-negligible effects.

Table 2. Normalized mutual information (NMI) between input features and resource dimensions on production traces. **Bold**: highest NMI per row.

	λ_t	λ_r	μ_{in}	σ^2_{in}	μ_{out}	σ^2_{out}
HBM	**0.63**	0.35	0.45	0.34	0.30	0.29
SM	**0.47**	0.36	0.42	0.33	0.32	0.32
BW	**0.48**	0.34	0.43	0.33	0.30	0.29

E Fragmentation Potential

For any vector $x \in \mathbb{R}^3$, define its centered version $x' \triangleq x - \frac{1}{3}\|x\|_1 \cdot \mathbf{1}$, which removes the mean component. The fragmentation potential is $\Phi(x) \triangleq \|x'\|_2^2 = \|x\|_2^2 - \frac{1}{3}\|x\|_1^2$. Geometrically, x' is the projection of x onto the zero-sum subspace orthogonal to $\mathbf{1}$, and $\Phi(x)$ is the squared norm of that projection. A useful property is $\Phi(S) = \Phi(r)$ where S is the bin's load and $r = \mathbf{1} - S$ its residual, since $\mathbf{1}' = \mathbf{0}$ implies $r' = -S'$.

Insertion rule. Inserting a model v into a bin with load S changes the potential by $\Delta\Phi = \Phi(v) - 2\langle r', v'\rangle$. Since $\Phi(v)$ is independent of the bin choice, minimizing $\Delta\Phi$ reduces to choosing $b^* = \arg\max_{b \in \mathcal{F}(v)} \langle r'_b, v'\rangle$, where $\mathcal{F}(v)$ is the set of feasible bins.

Damage ratio. The bin potential $\Phi(S)$ decomposes into an intrinsic part $\sum_i \Phi(v_i)$ (per-model anisotropy) and a cross-term capturing pairwise interactions. The damage ratio $\rho(B) \triangleq \Phi(r) / \sum_{i=1}^{k} \Phi(v_i)$ normalizes this: a low ρ indicates strong complementarity; ρ approaching 1 means the cross-term benefit has been lost.

F Workload Details

W0 (production trace). *W0* replays one full day of production traffic (March 10, 2026) covering 62 online applications and 9 models. Applications are grouped into three tensor-parallelism pools: TP1 (40 apps, 6 models), TP4 (16 apps, 2 models), and TP16 (6 apps, 1 model). Token rates stay below 0.5 M/min during the nighttime valley and rise to 5–6 M/min for head applications during the daytime peak. A large fraction of enterprise traffic is prefill-intensive [12]: classification, labeling, information extraction, and embedding tasks dominate, producing few output tokens per request.

W1 (production sample). *W1* samples representative head and tail applications from the same production environment and compresses their traffic into 120 s, with token rates scaled to the experimental cluster size.

W2–W7 (public datasets). *W2–W3* are derived from BurstGPT [58]: the single-model trace is split into multiple models with power-law–distributed token rates, compressed into 120 s. *W4–W5* are derived from LMSYS-Chat [77]: the dataset already contains multi-model request distributions; we compress the trace into 120 s. *W6–W7* are derived from ShareGPT [10]: the single-model token-length distribution is split into multiple models with power-law token rates, Poisson-distributed arrival timestamps, and time-varying request pressure, compressed into 120 s. All compressed workloads have their token rates scaled to match the experimental hardware capacity.